

Design and Development of Smart Charging Infrastructure for Electric Vehicles

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Abstract— Electric vehicles (EVs) are rising as one of the favourable strategies to overcome the increasing environmental concerns due to over-dependence on fossil fuels, emissions, and energy insufficiency. This work aims to design and develop a software framework to contribute to the EV charging infrastructure. The proposed work includes (1) creating front-end software for charging station operators and owners for managing and operating charging points, (2) developing cloud-based intelligent models for region prediction, price prediction, station filtering, and slot booking, (3) designing slot scheduling algorithms for optimal utilization of the charging station. Python and software tools like Streamlit, SQL, and Flask facilitate the implementation of the front-end software, database, and computational algorithms. The web application, built using the Streamlit Python framework, acts as the user interface for all stakeholders. The work utilizes the Linear Search Algorithm and the Binary Search Algorithm to filter stations based on the user requirements, such as distance and price range, which users indicate through the web application. The Gradient Boosting approach gives good results for region prediction for the chosen dataset of EV charging stations. In this work, a priority scheduling approach considering the time-of-use price achieves EV charging slot allocation. The software framework provides a comprehensive electric vehicle charging infrastructure where users can easily find the best charging station and reserve a spot for charging. The work's future extension can include the addition of functionalities for Vehicle-to-vehicle and Vehicle-to-grid.

Keywords— *Centralized Management System, Cloud-based solution, Electric Vehicle Charging Infrastructure, Electric Vehicles Charging scheduling, Electric Vehicle charging station selection, Electric Vehicles, Gradient Boosting, Streamlit.*

I. INTRODUCTION

In response to many climate challenges and global concerns about the increase in pollution from current fossil fuel-based vehicles on the road, there is significant focus on electrifying the transport sector as they provide a renewable-based solution, which is causing less pollution to the environment. This is pushing for a more extensive penetration of electric vehicles (EV), a revolutionary form of green transportation [1]. EVs have gained much acceptance in India, as they contribute substantially to reducing carbon dioxide emissions and fossil fuel consumption [2]. India is one of the few countries backing the global EV30@30 initiative, which aims to achieve at least 30 percent penetration of EVs in the automotive sector by 2030. However, India's journey towards electrification in transportation is still in its initial phases.

The essential components of the EV charging infrastructure are the Charging Management System (CMS), EV, and EVSE, and effective communication between these components is necessary for EV charging [5]. Numerous

studies explore how different communication technologies used in vehicles, such as cellular networks, ZigBee, Bluetooth, etc., facilitate the exchange of information among EVs, CMSs, and EVSEs. Also, [6] discuss a framework that combines charging slot reservations and parking time estimation by utilizing EVs predicted charging and arrival time. In terms of electric vehicles, each one will have unique charging needs, whether related to connectors, batteries, charging levels, or payment methods. Similarly, when considering a charging station, each EVSE will have its unique features [2]. Finding a convenient charging station for an owner is difficult, especially considering distance, payment, and time. Due to the unique characteristics of EVs and EVSEs, choosing the right charging station becomes challenging for EV owners. Even while the EV domain is witnessing many engineering solutions for the intelligent use of charging infrastructure, work that places a higher priority on EV user preferences is mainly absent. Gaining insights into the real-time charging station status can offer users valuable information, including the availability of charging facilities, reservations, and estimated time to reach the charging station [8].

The proposed solution in this paper includes front-end software, developed using Streamlit Python framework, for charging station operators and owners for managing and operating charging points, along with server-side and user-side software that can improve communication between EV users, charging stations, electrical utilities, and databases. The web framework includes a recommendation system that uses the user's location and any additional user requirements, such as connector type, state of charge (SoC), battery capacity, parking lot, accessibility, station category, weekday and access time, as data to suggest optimal charging station ranked by distance and price from the user's location. An intelligent model based on the Vincenty formula, Gradient Boosting Algorithm, Linear Search Algorithm, and Binary Search Algorithm achieves station filtering. The work also involves the selection of the best-fit machine learning algorithm based on various performance metrics for region prediction purposes.

The rest of the paper has the following sections. Section-II provides a summary of the previous work. In Section-III, an overview of the constructed system is available. Section-IV elaborates on the detailed implementation of the concept. Finally, the two sections, V and VI, give the result and conclusion.

II. LITERATURE SURVEY

The proposed project focuses on designing and developing Smart Charging Infrastructure for electric vehicles. Therefore, the literature analysis included EV and EVSE-related topics that help define the system development and feasibility verification of methodology. The review mainly focused on four areas: State of the art electric vehicle charging infrastructure, machine learning algorithms, cloud-based solutions in the EV domain, and scheduling. The following section thoroughly reviews various EV user requirements and vehicle parameters. [1] and [2] brings out different opportunities and challenges related to EVs. NITI AAYOG's "Handbook of Electric Vehicle Charging Infrastructure Implementation" presents a systematic approach to developing a roadmap for the implementation of EV charging infrastructure. It covers everything from evaluating EV charging needs to location selection, electricity supply arrangement, and on-ground implementation models [2].

E. ElGhanam et al. present a survey of existing vehicular communication technologies used to interact with EV, EVSE, and CMS [7]. This paper details ZigBee, Bluetooth, Radio frequency identification, and cellular networks and how they operate with the charging subsystem in EVs to exchange information like the SoC, user location, and charging decisions. The paper [8] presents the development of CMS software that helps charge EVs. CMS includes an Android Studio-created mobile application, a web-based application created using HTML and PHP, acting as the front-end, and connecting to a MySQL database through a local server. ESP8266 Wi-Fi modem, Arduino, and controller area network (CAN) module realized the communication capabilities of EVSE and EV.

[10] explores various deep neural networks and supervised and unsupervised machine learning techniques for forecasting EV charging behavior utilizing different parameters like arrival time, departure time, CPU type, etc. These methods address traffic prediction, routing and classification, congestion control, resource allocation, and fault detection. The study in [12] utilizes the Support Vector Machine (SVM) machine learning method to analyse the home charge schedule. The SVM method accurately predicts the electric car's condition, such as idle, grid-to-vehicle (G2V), or vehicle-to-grid (V2G), with approximately 100 percent accuracy. [13] describes a clever program that enables users to look at various tariff prices on the grid. Tariff prices cover charges for supplying and utilizing energy to the grid. [7] offers a server-based forecasting program made to manage schedules and reduce waiting times. Additionally, it offers real-time recommendations for charging stations based on the shortest travel distance and fastest charging times. [15] suggesting a shift from the conventional control setup of a battery system to an intelligent battery management and control system linked with cloud computing. [16] emphasis is on creating a cloud-oriented system for recommending EV charging stations. This system offers users suggestions for suitable charging stations based on location and requirements. The system will use the user's location and requirements as inputs to rank the closest charging stations to

the user's location in order of convenience and to make slot reservations.

[18] proposes an intelligent solution to oversee grid-connected EV charging stations powered by photovoltaic sources. This solution comprises a mobile application enabling EV drivers to reserve charging slots, an optimization algorithm for scheduling charging sessions, and a remote execution module for managing charging procedures via the open charge point protocol (OCPP).

The majority of recent research places minimal attention on extra user requirements or preferences and instead prioritizes selecting charging stations that are as cheap and as close as feasible. Therefore, the proposed effort focuses on creating an intelligent EV charging infrastructure to assist EV users in choosing the charging station that best suits their needs. The earlier efforts deal with energy consumption monitoring, scheduling or cancelling charging slots, identifying charging stations along the trip, and figuring out where in the line-up to charge. Our research, however, focuses on the possibilities of one application architecture that would include all of these services and capabilities.

III. SYSTEM OVERVIEW

Figure 1 provides an overview of the system employed in this project. The various stakeholders have a web application that serves as the user interface and has functions like adding EV user, EVSE, station filter, slot book, etc. The system's user interface enables an EV user to specify the station's preference. The streamlit cloud, connected to the charging station database, receives charging requests and user specifications. Blocks of the station selection model use the station database to implement algorithms for region prediction, distance calculation, pricing prediction, station filtering, and slot booking and provide station data to the user interface. As a result, EV users may quickly locate nearby charging stations that meet their criteria. The user may reserve a time slot after selecting the appropriate station from the list, and the server will immediately begin the scheduling process. Additionally, the web-based application will display all the information.

The Streamlit framework is the tool used to design the user interface, while MySQL is the database management system used to store the data. Based on these user requirements, the station selection model does area clustering using a KNN model and region prediction using a gradient boosting model. A subset of stations near the vehicle's position is filtered using a clustering and region prediction model, which partitions the whole dataset into groups of locations nearby. This subset is then passed to the pricing prediction model using random forest regression and the distance calculation model using the Vincenty formula. The station filtering model then uses linear and binary search algorithms to exclude stations that do not fulfill user demands. The final sorting to filter the stations accommodates user-specified distance and price range.

Users can view the list of stations and their locations displayed on Mapbox maps in the Streamlit cloud framework. Then, using the Flask framework and HTTP method, a user can submit a booking request with helpful information for a specific station ID to the server side. The priority scheduling approach used facilitates server-side.

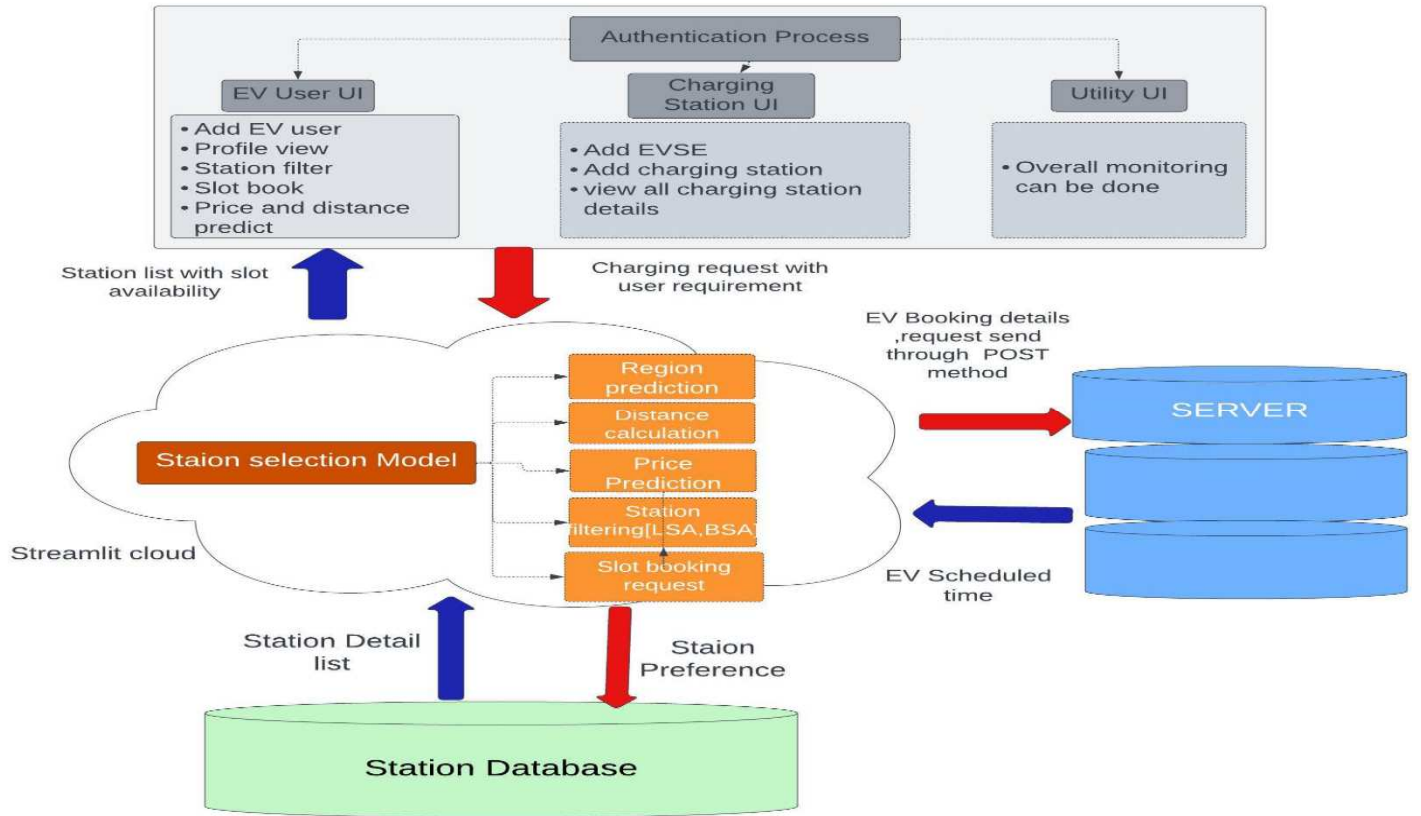


Fig. 1: System Architecture of Smart Charging Infrastructure of Electric Vehicle

IV. IMPLEMENTATION

The initial phase of this project involves the generation of a comprehensive dataset for EV charging stations, encompassing all essential information. The dataset containing details of 24,740 US EV charging stations was retrieved using the GET method, and generated a CSV file with the response content to form the database for the project [20]. A dataset from Harvard Dataverse with data from 3,395 high-resolution electric car charging sessions for the pricing prediction is also used [21]. Merging these two datasets produces a larger dataset that will serve as a more precise database for the intended task.

A. Authentication and User Interface

The UI of the web application has login, register, and dashboard pages that different users may view for authentication purposes. Using authentication, EV users can sign in to their accounts, and each user is assigned a unique user ID to carry out the rest of the booking procedure. An EV user can select their preferred station specifications using the system's UI. A MySQL database stores all credential information. Additionally, the UI allows for the addition of vehicles by EV users, EVSEs by CS administrators, and CS by electrical utility administrators. Streamlit, a Python framework for constructing unique online apps, is utilized for UI development. After being created, the station selection model's Streamlit application launched into the Streamlit Cloud uses various Streamlit APIs.

B. Region Prediction

Train-test dataset separation follows the dataset pre-processing. The KNN algorithm used a clustering approach to group neighbouring stations into regions and trained each area as a specific class. Figure 2 shows how the dataset is clustered using KNN and then assign classes to each cluster to determine the user's location cluster. Different machine learning algorithms, including Support vector machines (SVM), the Naive Bayes model, the Random Forest algorithm, and the Gradient Boosting model, serve for multi-class classification. Several performance measures, such as confusion metrics and classification reports, bring out these algorithms' performance.

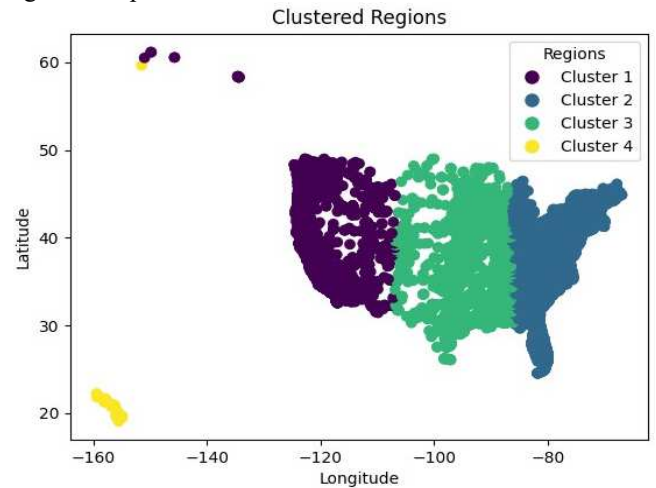


Fig. 2: Clustered Region

Based on performance evaluations, the gradient-boosting approach makes region prediction. Gradient Boosting is an effective machine learning method that is a member of the collaborative learning family. It integrates several weak predictive models (usually decision trees) to construct a robust predictive model. The algorithm operates by repeatedly adding new models to fix mistakes produced by earlier versions. The model takes the user location as input for the prediction of labels. Based on these labels, the region extraction happens once the model training is complete using latitude, longitude, and labels.

C. Station Filtering

In this stage, the parameters considered include user needs such as the connector type, distance, weekday, station category, accessibility, SoC, battery capacity, access time, and parking lot requirements. After region prediction, the subset of stations obtained undergoes a content-driven filtering procedure using the Linear Search Algorithm. Following this, the system calculates and predicts the distance and price for the filtered stations and then uses a binary search algorithm to locate the station that will satisfy the user-specified range. In LSA, the algorithm evaluates individual station content sequentially and upon identifying a match, adds the distinct data of the station into a data structure. Following that, the binary search algorithm executes on the data frame. Figure 3 illustrates the BSA flowchart's operation within the model. The result is a specific station group that meets the user's needs.

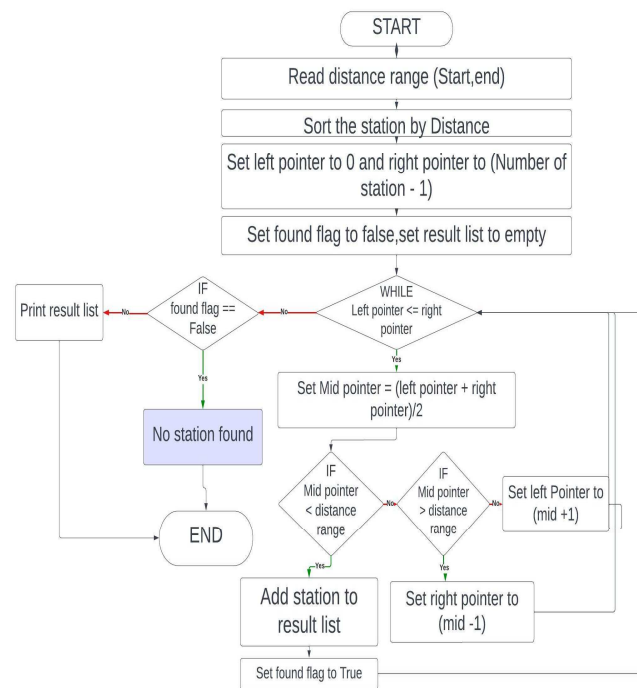


Fig. 3: Binary Search Algorithm

D. Distance Calculation

Distance calculations depend heavily on latitude and longitude. Station filtering gives a data frame with the information of the filtered station which satisfies the user requirements. Using the Vincenty formula, each station's location-related data takes into account this data frame and determine the distance from the location of the EV user. Then

the system will rank the stations by increasing the distance from the vehicle after calculating the distances.

E. Scheduling

The infrastructure for electric vehicles prioritizes the charging of EVs at public charging stations based on a priority schedule. This way, high-priority EVs can charge first, while lower-priority EVs may have to wait. Utilizing some variables, such as the SoC of the EV, its arrival time, and its destination, determines its priority. EVs travelling to vital locations, such as medical appointments, may have a higher priority. Incorporating the concept of weight into the priority scheduling allows for determining each user's priority based on their preferences. Depending on their choices and priorities users may change the weights allocated to variables like cost, waiting time, and distance. For instance, a client might give more weight to the cost aspect in the priority calculation if they prefer cost savings over waiting time. On the other hand, if minimising waiting time is crucial, then the user might give more significance to that component.

V. RESULTS AND DISCUSSION

Figure 4 displays the authentication page created using Streamlit. The authentication page features a user type option that allows users to choose from various stakeholders, including an EV user, a charging station administrator, or a utility administrator. After selecting, users can log in or sign up for the app by entering their username and password.

Table 1: Model Performance Comparison

Model Name	Accuracy	Precision	Recall	F1-Score
Random Forest	0.99	0.99	1.00	1.00
SVM	0.96	0.97	0.97	0.97
Gradient Boosting	1.00	1.00	1.00	1.00
Naive Bayes	0.91	0.93	0.92	0.92

Home Page

SELECT THE USER TYPE

EV user

EV user

CS admin

Utility admin

Login Section for EV user

Username

Password

☐ Login

Fig. 4: Home page

Table 1 shows the performance study of machine learning methods utilizing the confusion matrix and classification report. When comparing the confusion matrix and precision, recall, and f1-score metrics derived from the classification reports of SVM, RFA, Gradient Boosting, and Naive Bayes methods, it becomes evident that the Gradient Boosting

approach is better suited for this classification issue involving multiple classes. Users enter their information and needs on the application's user page, visible in Figure. The application displays the result obtained after sorting, measuring distances, and filtering stations in a tabular format. Consider an instance where the consumer requires open, public charging stations with Tesla connections. The user is also searching for stations that resemble hotels and remain open 24 hours a day on Monday. If the user requests a specified distance and price in addition to other conditions, the station will undergo filtering. After providing the SOC and the vehicle's battery capacity details, the user can click "Search Station" to find the closest EV station that matches the requirements. Figure 5 shows the search results, when the user provides the necessary information. The software presents a compilation of stations, featuring their accessible slots, waiting queue, power, distance, and pricing.

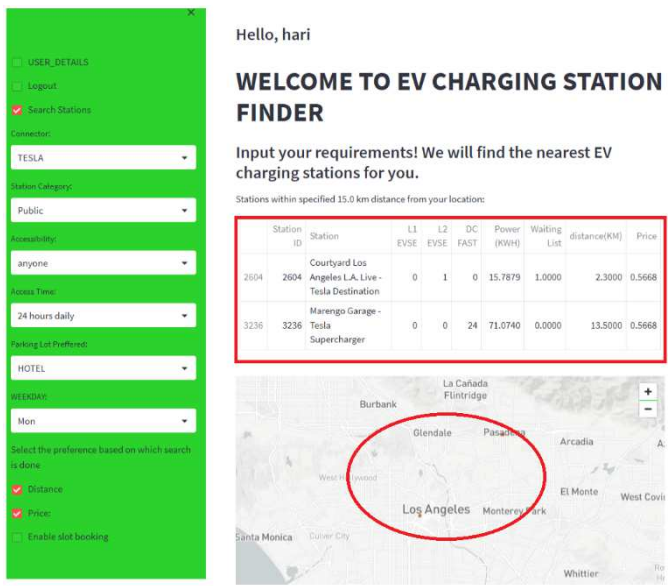


Fig. 5: Outcome of a sample for specific user input

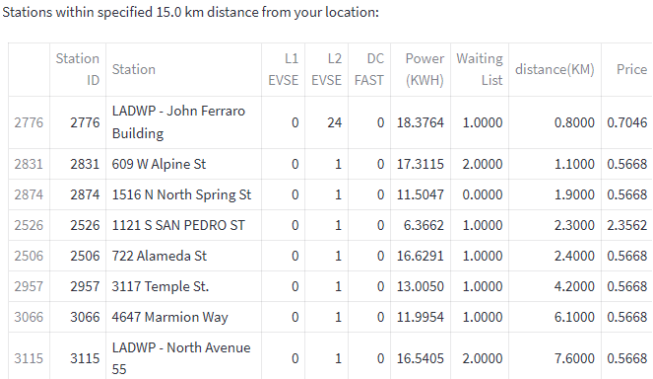


Fig. 6: Sampled filtered Station

When locating a charging station that aligns with their requirements, the user must then submit a booking request with various details, including Station ID, preferred time range (to consider the time of use pricing), distance, price, start time, end time, various weights for priority calculation, and the type of EVSE they need. Figure 7 shows the process of sending a slot booking request from the application to the server, where all EV scheduling occurs. Sending a booking request to the server, where all user information is gathered and organized according to station ID and time, will result in

a successful booking if a free space is available for that EVSE. The reservation is successful if only one EV attempts to book at the correct time and charging station. The priority-based scheduling approach addresses the issue when multiple EVs try to access the same station simultaneously.



Fig. 7: Waiting list indicated

The server-side utilizes a dictionary structure to store all EV data (as shown in Figure 9) and then sorts it based on station ID and time. Regarding multiple EV reservations for the same time slot, priority allocation consider factors such as SoC, arrival time, charging time, distance, time of use price, and various weights. The scheduling of EVs for different available slots happens based on their priority. When an EV user tries to reserve a spot with an existing waiting list, the details get added to the waiting list, as shown in Figure 7. A notification reading "No available slots" will be displayed if no additional slots are vacant.

Charging Station ID: 3236
Time slot : 9:00 - 12:00
Charging process started.
Charging EV2-hareshma ...
Charging EV2-hareshma :20/20 minutes complete.
Charging EV1-Sathi ...
Charging EV1-Sathi :30/30 minutes complete.
Charging EV4-Raj ...
Charging EV4-Raj :10/10 minutes complete.
Charging EV3-Anjali ...
Charging EV3-Anjali :40/40 minutes complete.
Charging process complete .

Fig. 8: Priority scheduling on the server side

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hari
{1: {'soc': 30, 'station_id': 2874, 'Start_time': '08:15:00', 'End_time': '08:15:00', 'Cost': 0.56, 'TOP_range': '06:00-18:00', 'charging_slot': 1, 'distance': 1.11, 'ev_user':
127.0.0.1 - - [19/Dec/2023 08:15:38] "POST /booking HTTP/1.1" 200 -
```

Fig. 9: Booking request received in server from the web application

VI. CONCLUSION

The intended work aims to create and construct an innovative EV charging infrastructure that enables EV users to choose charging stations based on their comfort and preferences. Previous works actively addressed aspects such as monitoring energy consumption, arranging or discontinuing charging slots, identifying charging stations along the travel route, and establishing the positioning in the queue at the charging station. However, our study focuses on the feasibility of a unified software framework that integrates all these services and capabilities. The station selection model gathers user input from the user interface and performs activities such as region prediction, station filtering, distance computation, pricing prediction, and station sorting. It lists suitable stations and details their pricing, distance, power supply, and slot availability. Future enhancements may involve updating the application's station list using live locations by incorporating popular standards and protocols like Open Charge Point Protocol, Open Charge Point Interface protocol, and Open Clearing House Protocol. Additionally, the infrastructure can have features that integrate V2V and V2G.

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